Ames Housing Price Prediction

Final report: regression workflow on 1,460 house sales (80 features, target = Sale_Price)

1. Objective and Data

The goal is to predict the sale price of a house from its physical and location features. The dataset has 1,460 sales with 37 numerical and 43 categorical columns; no duplicate rows. Sale price is right-skewed (mean $180,921, median $163,000, range $34,900 to $755,000, SD $79,443). Four columns were dropped for being almost empty (Pool_Quality 99.5% missing, Miscellaneous_Feature 96.3%, Alley_Type 93.8%, Fence_Quality 80.8%). In most other columns a missing value means the feature does not exist (no garage, no basement, no fireplace), so these were filled with a "Not Available" category rather than treated as unknown.

Column	Missing	Handling
Pool_Quality, Misc_Feature, Alley, Fence	80.8% - 99.5%	Dropped
Masonry_Veneer_Type	59.7%	"Not Available" (no veneer)
Fireplace_Quality	47.3%	"Not Available" (no fireplace)
Lot_Frontage_Feet	17.7%	Median imputation
Garage columns (5)	5.6%	"Not Available"; Garage_Year_Built = Year_Built
Basement columns (5)	2.5% - 2.6%	"Not Available" (no basement)
Electrical_System, Masonry_Veneer_Area	0.07%, 0.55%	Mode, 0

Table 1. Missing values (19 columns affected) and how each was handled.

2. Key Findings from Exploratory Analysis

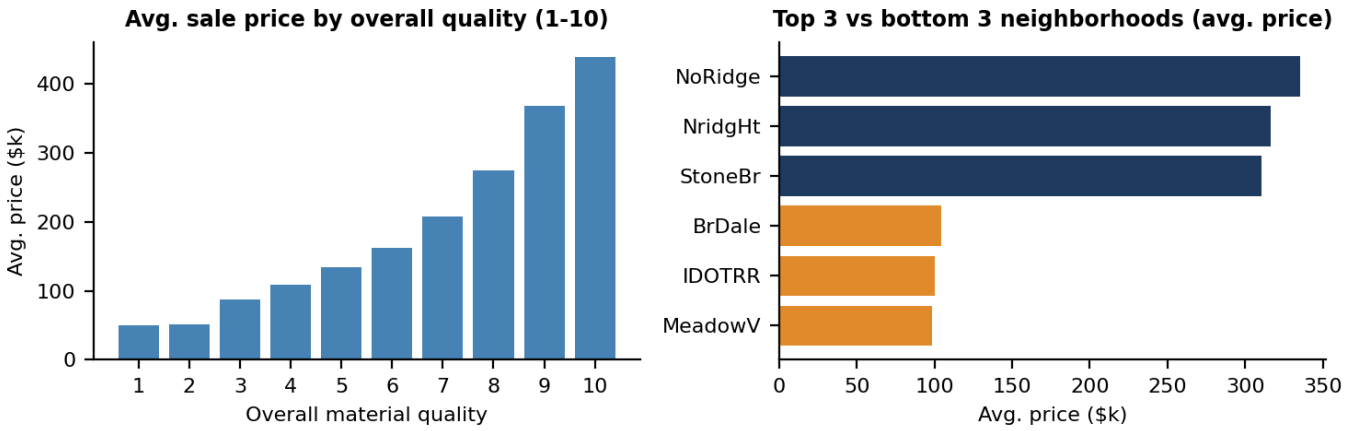
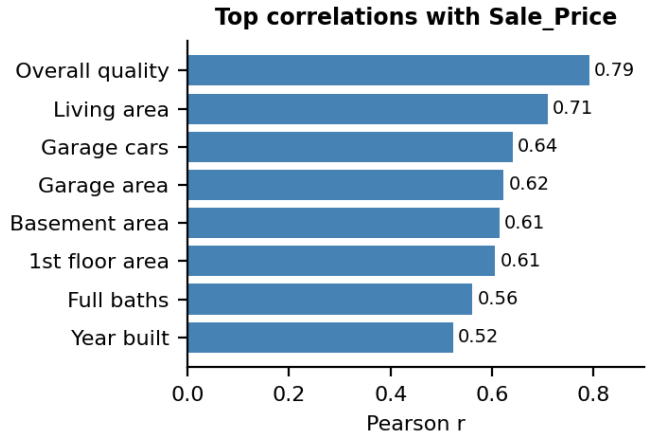


Figure 1. Price rises steeply with quality (about $50k at quality 1-2 to $439k at quality 10); the best neighborhoods average 3.4x the cheapest.



- **Quality and size dominate.** Overall quality (r = 0.79) and above-grade living area (r = 0.71) are the strongest numeric correlates.
- **Location matters.** NoRidge ($335k), NridgHt ($316k) and StoneBr ($310k) are the dearest; MeadowV ($99k), IDOTRR ($100k) and BrDale ($104k) the cheapest.
- **Amenities carry a premium.** Central air +$80,923; basement +$77,225; kitchen quality Excellent vs Typical +$188,592; 3-car garage averages $309,636 vs $183,852 for 2-car.
- **Age lowers price** (house age vs price r = -0.52). Seasonality is mild: best month Sept ($195.7k), worst April ($171.5k).
- **Remodeling does not help on average**: remodeled houses average $179,096 vs $182,584 for non-remodeled (-$3,487).

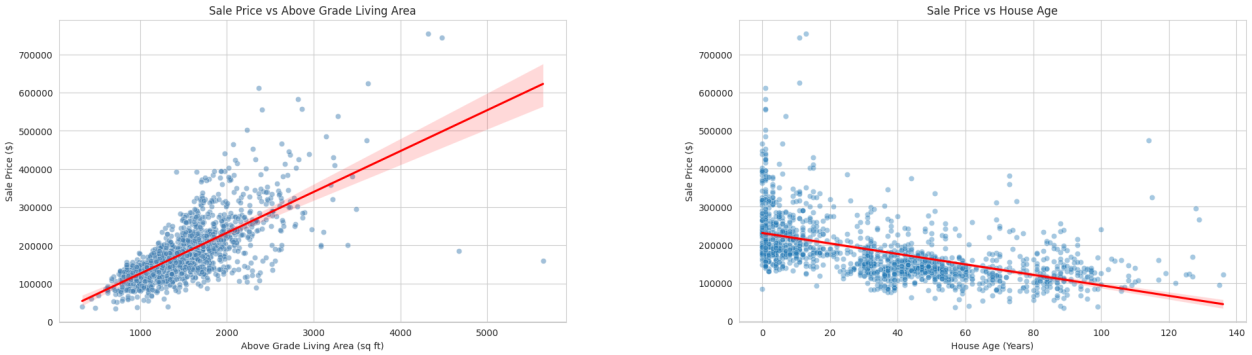


Figure 2. Left: price vs above-grade living area (r = 0.71). Right: price vs house age (r = -0.52). Both show clear trends with a few very large, very expensive outliers, which motivated RobustScaler and the log target.

3. Preprocessing and Pipeline Design

The data was split 64% train (934), 16% validation (234) and 20% test (292) with a fixed seed (42). All preprocessing sits inside a scikit-learn *Pipeline* and *ColumnTransformer*, so every statistic is learned on training data only (no leakage):

- **Numerical (37):** median imputation, Yeo-Johnson PowerTransformer for skew, RobustScaler (robust to outliers such as very large houses).
- **Nominal (24):** constant imputation, then one-hot encoding. **Ordinal (19):** ordered encoding (e.g. Po < Fa < TA < Gd < Ex).
- **Target:** modeled as log(1 + price) via TransformedTargetRegressor; predictions are converted back to dollars, so all metrics are in dollars.
- **Feature selection:** SVR and KNN pipelines add Lasso-based SelectFromModel (106 features kept); the baseline Linear Regression uses all features.

4. Models and Results

Three model families were compared: Linear Regression (baseline), Support Vector Regression (SVR) and K-Nearest Neighbors (KNN). SVR and KNN were each run with default settings and with a GridSearchCV (5-fold, training data only) over C, kernel, gamma, degree (SVR) and k, weights, p (KNN).

Model	Grid searched	Selected
SVR	C {0.1, 1, 10, 100}; kernel {rbf, poly}; gamma {scale, 0.01, 0.001}; degree {2, 3}	C=0.1, rbf, gamma=scale (CV score nan)
KNN	k {3, 5, 7, 9, 11, 15}; weights {uniform, distance}; p {1, 2}	k=3, uniform, p=1 (CV score nan)

Table 2. Hyperparameter search setup (GridSearchCV, 5-fold, RMSE scoring).

Model	Val MAE	Val RMSE	Val R^2	Test RMSE	Test R^2	Train-Val R^2 gap
Linear Regression	**$15,491**	**$24,140**	**0.910**	**$24,768**	**0.920**	**0.035**
SVR (default)	$16,858	$26,114	0.895	$29,658	0.885	0.043
SVR (tuned)	$19,592	$31,305	0.848	$36,634	0.825	0.012
KNN (default)	$22,607	$36,060	0.799	$36,565	0.826	0.046
KNN (tuned)	$23,174	$35,898	0.801	$35,999	0.831	0.092

Table 3. Metrics in dollars (MAE, RMSE) and R^2. Best model highlighted.

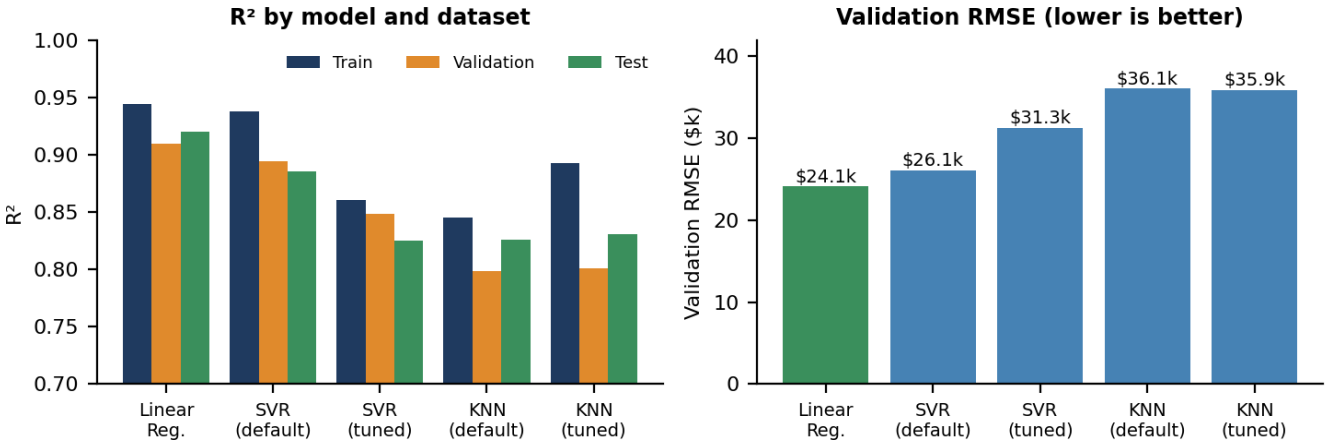


Figure 3. Linear Regression leads on every validation and test metric; KNN trails clearly.

5. Evaluation and Interpretation

- **Best performer:** Linear Regression has the lowest validation RMSE ($24.1k) and the lowest test RMSE ($24.8k), with R^2 of 0.91 (validation) and 0.92 (test). After log-transforming the target and encoding features carefully, a simple linear model is hard to beat on this dataset.
- **Fit quality:** the train-validation R^2 gap is small for all models (0.01-0.09), so none overfits severely. KNN is the weakest (validation R^2 about 0.80): with more than 200 columns after one-hot encoding, distance-based neighbours are not truly similar houses. Tuned KNN shows the largest gap (0.092), the clearest overfitting signal.
- **Tuning did not help.** The tuned SVR (C = 0.1, rbf, gamma = scale) is worse than the default SVR (validation RMSE $31.3k vs $26.1k). The notebook reports the best CV RMSE as *nan* for both grid searches, which means the cross-validation scores were not computed, so the "best" parameters are effectively just the first combination in each grid and should not be trusted.

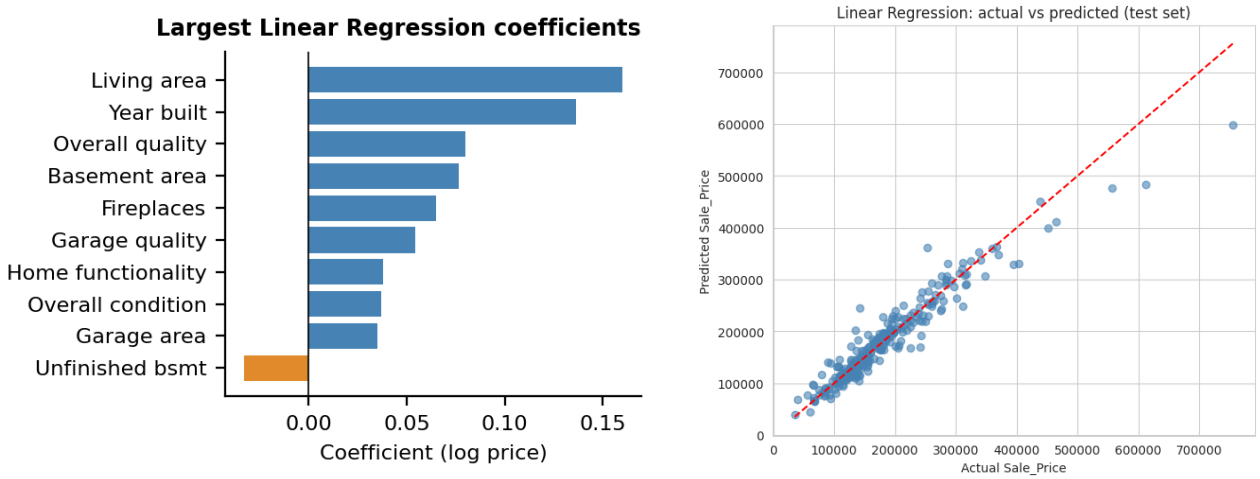


Figure 4. Left: largest Linear Regression coefficients (numerical and ordinal features only). Right: actual vs predicted prices on the test set; points hug the diagonal, with the largest errors on very expensive houses.

6. Final Model Selection

Selected model: Linear Regression (log-price target, full preprocessing pipeline). It has the best validation RMSE ($24,140, R^2 0.910) and the best test result (RMSE $24,768, MAE $15,446, R^2 0.920), a modest generalization gap (0.035), and is the easiest model to explain. Its largest drivers agree with the exploratory analysis: living area, year built, overall quality, basement size and fireplaces all push predicted price up.

Limitations and next steps. (1) Fix the hyperparameter search (check why CV scores are nan, e.g. run with error_score="raise") and re-run before drawing conclusions about SVR and KNN. (2) The test set is small (292 houses), so results carry some uncertainty; repeated cross-validation would be more stable. (3) Errors grow for the most expensive homes; tree ensembles or regularized linear models (Ridge/Lasso) are natural candidates to try.